Corporate Project Management System

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SUBJECT :- DBMS

SEM :-6

Scope & Entities

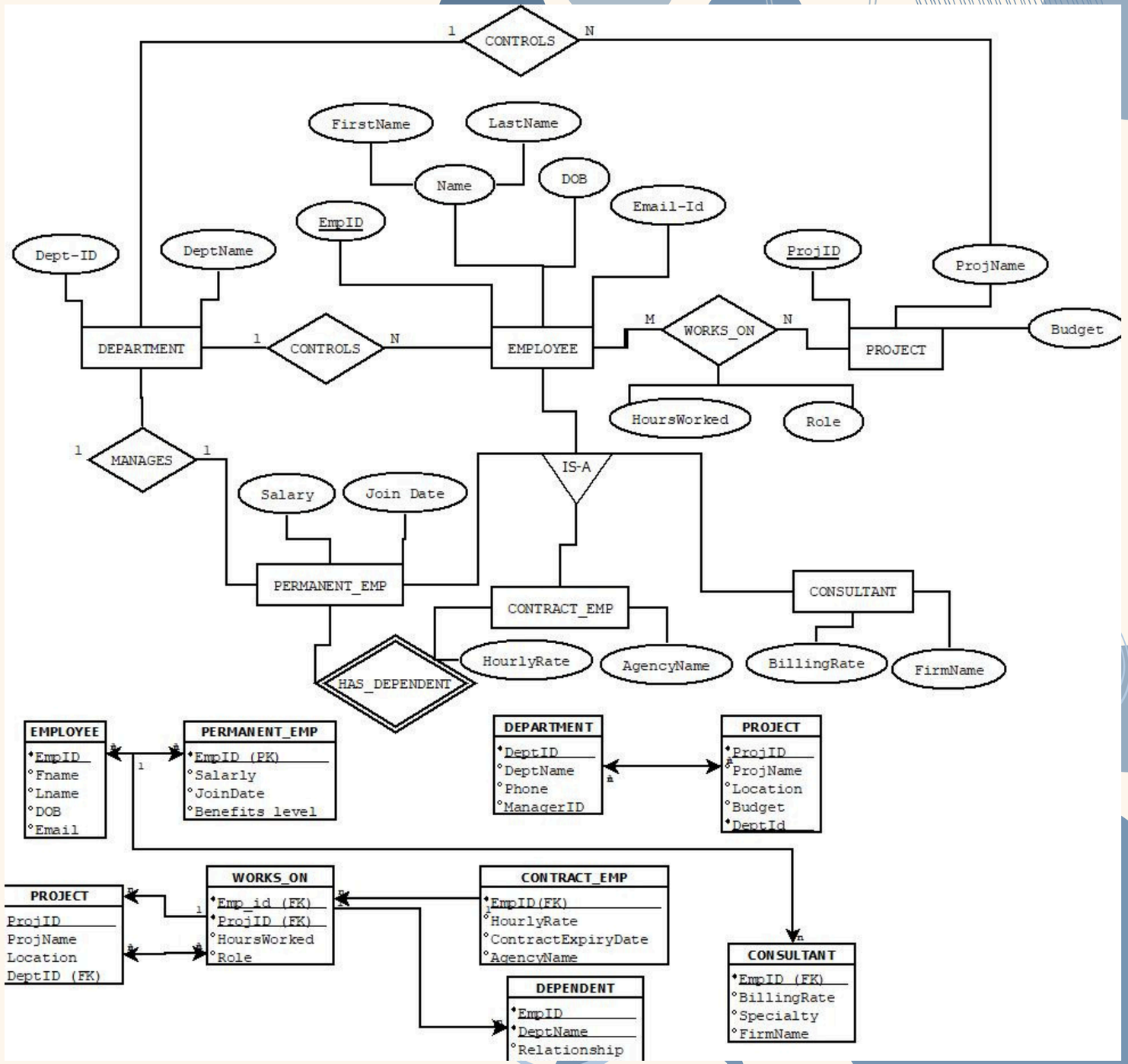
- Core Entities: Employee, Department, Project.
- Specialization (Hierarchy):
- Superclass: Employee (General data).
- Subclasses: Permanent Employee, Contract Employee, Consultant.
- Weak Entity: Dependent (Relies on Permanent Employee).

The ER/EER Diagram Structure

- Cardinalities:
- 1:1 between Manager and Department.
- 1:N between Department and Project.
- M:N between Employee and Project (Works_On).
- Participation Constraints:
- Total Participation: Projects must have a Department; Dependents must have an Employee.
- Partial Participation: Not all Employees are Managers.

Relationship	Connected Entities	Cardinality	Business Rule
MANAGES	Perm. Emp ↔ Dept	**1:1**	One manager per department.
CONTROLS	Dept ↔ Project	**1:N**	One dept controls many projects.
WORKS_ON	Employee ↔ Project	**M:N**	Employees work on multiple projects.
HAS_DEP	Perm. Emp ↔ Dependent	**1:N**	One employee has many dependents.

ER/EER Diagram Structure



Specialization & Generalization

- Strategy: Disjoint, Total Participation.
- Logic: An employee must be exactly one type (Permanent, Contract, OR Consultant).
- Attributes:
- Permanent: Salary, Benefits.
- Contract: Hourly Rate, Agency.
- Consultant: Billing Rate, Firm Name.

Mapping to Relational Model

- Step 1: Create table for Superclass EMPLOYEE.
- Step 2: Create separate tables for Subclasses (PERMANENT, CONTRACT, CONSULTANT) linked by EmpID (Primary Key).
- Step 3: Map M:N relationship WORKS_ON to a dedicated table (Composite Key).
- Step 4: Map 1:N relationship CONTROLS by adding DeptID FK to PROJECT table.
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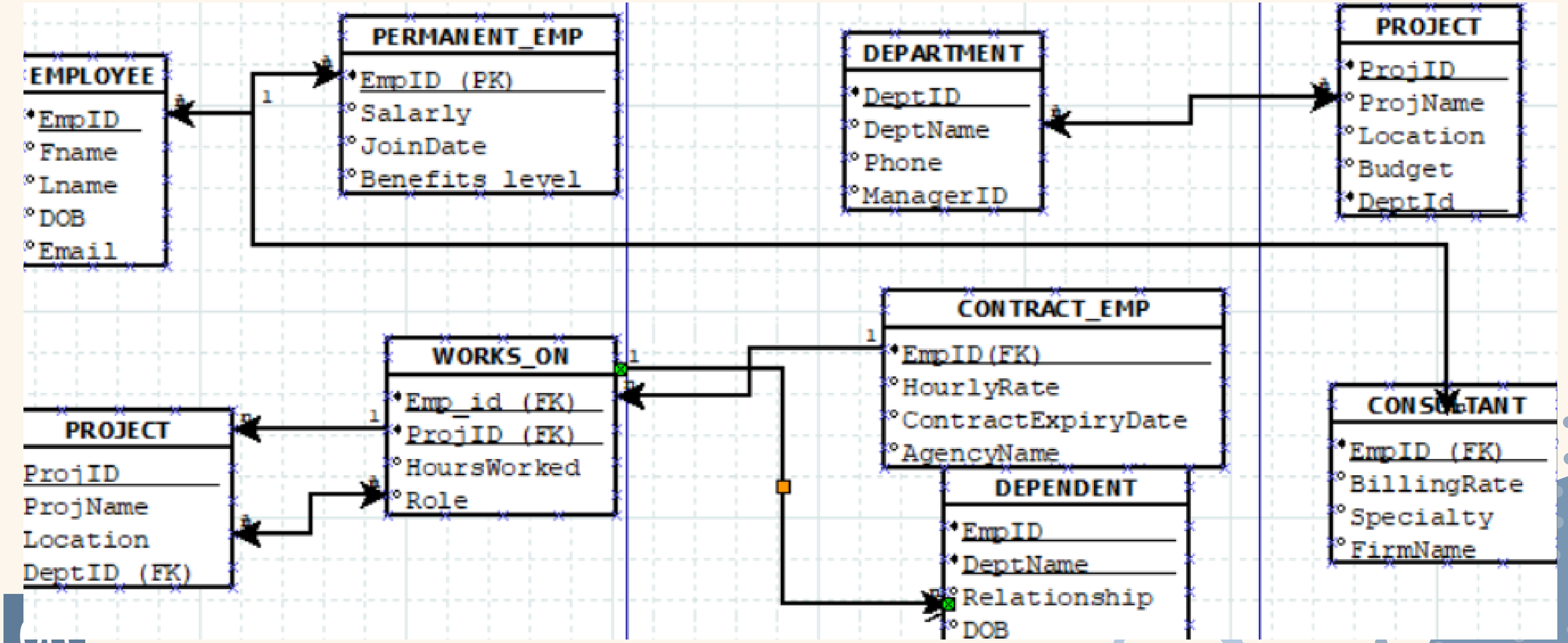
Key Constraints

- Primary Keys (PK): Uniquely identify rows (e.g., EmpID, ProjID).
- Foreign Keys (FK): Enforce links (e.g., ManagerID links Department to Employee).
- Domain Constraints: Salary > 0, Email must be unique.
- Semantic Constraint: Only Permanent Employees can manage Departments.

Mapping Strategy (ER to Relational)

- Mapping Specialization: Created 4 tables. EMPLOYEE (Parent) + 3 Child tables (PERMANENT, CONTRACT, CONSULTANT). Linked via EmpID.
- Mapping 1:N (CONTROLS): Added DeptID as a Foreign Key in the PROJECT table.
- Mapping M:N (WORKS_ON): Created a junction table WORKS_ON with Composite Key (EmpID, ProjID).
- Mapping Weak Entity: DEPENDENT table uses Composite Key (EmpID, DepName) and ON DELETE CASCADE.

Relational Schema (The Tables)



Summary & Conclusion

Summary:

- Modeled complex workforce hierarchy using EER Specialization.
- Enforced Referential Integrity and business rules via constraints.
- Normalized design ensures minimal data redundancy.

Conclusion:

- The system is scalable (easy to add new employee types).
- The "Table-per-Class" strategy balances performance and data integrity.
- Ready for SQL implementation.

Thank you